Resonance in Shaking Device

How does power dissipation in a shaking device respond to changes in the spring constant caused by altering the length of the stainless steel rod (connection between actuator and load)?

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I. Introduction

As a physics student interested in mechanical engineering, I was interested in resonance when studying waves. The spring-mass-damper system is a fundamental concept in mechanical engineering, which further inspired me to explore a research topic that connects these two ideas.

This study investigates the research question, "*How does power dissipation in a shaking device respond to changes in the spring constant caused by altering the length of the stainless steel rod (connection between actuator and load)?*" By exploring the relationship between power consumption and spring constant, this study aims to offer insights into designing more efficient systems and improving power consumption in vibrational applications.

II. Background

A. Simple Harmonic Motion

The simple harmonic motion (SHM) can be defined as a motion in which the force's magnitude should be proportional to the object's displacement from its equilibrium point. The following equation describes the displacement of SHM:

$$x(t) = X_0 \sin(\omega t + \phi), \tag{1}$$

where displacement x can be written in a sine function of time t, X_0 is the amplitude, ω is the angular frequency, and ϕ is defined as the phase angle. Since acceleration is the second derivative of displacement with respect to time, equation (1) gives:

$$a(t) = -\omega^2 x(t). \tag{2}$$

That is, the acceleration is proportional to the object's displacement from the equilibrium point in SHM.

B. Spring-Mass-Damper System

In a mass-spring-damper system, let m, d, and k respectively denote mass, viscous damping coefficient, and spring constant. The following differential equation describes the motion of the mass-spring-damper system:

$$m\frac{d^2x}{dt^2} + d\frac{dx}{dt} + kx = F(t), \text{ or} \tag{3}$$

$$\frac{d^2x}{dt^2}+2\xi\omega_n\frac{dx}{dt}+\omega_n^2x=f(t), \tag{4}$$

where $\omega_n=\sqrt{\frac{k}{m}}$ is the undamped natural frequency, $\xi=\frac{d}{2\omega_n m}$ is the damping ratio, $f(t)=\frac{F(t)}{m}$. The solution of equation (4) is supposed to be the form of $x(t)=Ae^{\lambda t}$, in which A and λ are constants. Equation (4) gives the following equations with a variable, λ:

$$\lambda^2+2\xi\omega_n\lambda+\omega_n^2=0, \tag{5}$$

$$\Rightarrow \lambda=\frac{-2\xi\omega_n\pm\sqrt{4\xi^2\omega_n^2-4\omega_n^2}}{2}=-\xi\omega_n\pm\omega_n\sqrt{\xi^2-1}. \tag{6}$$

Let $x(0)=0, \dot{x}(0)=0$, and $f(t)=\omega_n^2$, in which ω_n is constant. Consider three cases as follows:

1) **Overdamped** case ($\xi>1$):

$$x(t)=1+\frac{\omega_n}{2\sqrt{\xi^2-1}}\left(\frac{e^{-\lambda_1 t}}{\lambda_1}-\frac{e^{-\lambda_2 t}}{\lambda_2}\right), \tag{7}$$

where $\lambda_1=\omega_n\left(\xi+\sqrt{\xi^2-1}\right), \lambda_2=\omega_n\left(\xi-\sqrt{\xi^2-1}\right)$.

2) **Critically Damped** case ($\xi=1$):

$$x(t)=1-e^{-\omega_n t}\left(1+\omega_n t\right). \tag{8}$$

3) **Underdamped** case ($0<\xi<1$):

$$x(t)=1-e^{-\xi\omega_n t}\left(\cos\omega_d t+\frac{\xi}{\sqrt{1-\xi^2}}\sin\omega_d t\right), \tag{9}$$

where $\omega_d=\omega_n\sqrt{1-\xi^2}$ is the damped natural frequency. Thus, the oscillation frequency of an underdamped system is $\omega_d=\frac{\left(4km-d^2\right)^{\frac{1}{2}}}{2m}>0$.

These equations lay as the framework of the **Theoretical Analysis** section.

C. Spring Constant

In the shaking device, a rotary actuator is oscillating in both clockwise and counterclockwise directions. For the further analysis of the system, the torsional spring constant of stainless steel rods is crucial. The spring constant of a straight wire can be expressed by the following equation (Liu et al., 2016):

$$K = \frac{G\pi d^4}{32l}, \tag{10}$$

where G is the shear modulus, d is the diameter of the torsion wire, and l is the length of the torsion wire.

Young's modulus is essential for calculating the shear modulus. The relationship between shear modulus and Young's modulus is (Uday Chippada et al., 2010):

$$E = 2G(1+v), \tag{11}$$

where E is the Young's modulus and v is the Poisson's ratio. The constant K is part of the independent variable as the value of the K changes as the independent variable, length of the stainless steel rod, alternates.

D. Vibration

Vibration is a periodic response of a mechanical system about an equilibrium position, with its rate defined as the term "frequency." In mechanical systems, vibrations can occur in free or forced vibrations. Forced vibration is induced through excitation (De Silva, C.W., 2006). These excitation forces can be transmitted to a system from external or internal sources.

E. Resonance

Resonance happens when the frequency of an external force is the same as the system's resonant frequency, resulting in an increased amplitude (De Silva, C.W., 2006). While it relates to quasi-periodical behaviors, resonance can be undesirable to the consequences. For example, resonance caused the collapse of the Tacoma Narrows Bridge, USA, which is an example of bridge failure (Choudhury J.R., Hasnat A., 2015). Therefore, desired vibrations are the general goal of vibration engineering.

F. Shaking Device

Shaking devices have many practical applications, such as accelerating vial shaking process after drug reconstitution or for automatic chemical liquid shaking (Ding et al., 2020). Another example is shaking bioreactors, the most frequently used system for process optimization (Klöckner, W., & Büchs, J., 2012). Due to the ubiquity of shaking devices, reducing actuators' power loss is meaningful. While resonance is typically destructive, this study investigates its potential to reduce power consumption in shaking devices.

III. Methodology

A. Theoretical Analysis

A schematic of a conventional shaking device is shown in **Fig. 1**, where the actuator is firmly attached to the load, allowing it to oscillate in both clockwise and counterclockwise directions. The same applies to the following schematics and the experiment.

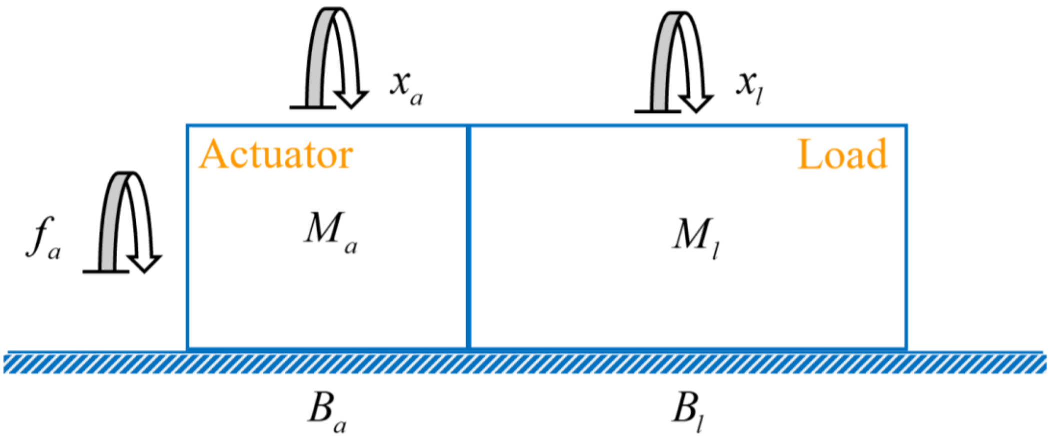

Fig. 1. Schematic of a conventional shaking device

Here, f_a denotes the torque produced by the actuator, M_a, and B_a respectively denote the actuator's moment of inertia and viscosity coefficient. Likewise, M_l and B_l respectively represent the load's moment of inertia and viscosity coefficient. In this case, the load's angular position, x_a, is the same as the actuator's position, x_l. Let the desired motion of the load be specified by:

$$x_l = X_d \sin(\omega_d t), \tag{12}$$

in which X_d and ω_d respectively denote the desired amplitude and frequency. In this configuration of rigid connection, the actuator and the load have the same motion. Since $x_a = x_l$, one has $\dot{x}_a = \dot{x}_l$. The instantaneous power dissipated by viscous friction at the actuator side is $P_r = B_a \dot{x}_a \dot{x}_a = B_a \dot{x}_l \dot{x}_l = B_a X_d^2 \omega_d^2 \cos^2(\omega_d t)$, which gives the average power:

$$P_r^{\text{avg}} = \frac{1}{2} B_a X_d^2 \omega_d^2. \tag{13}$$

The idea is to introduce a flexible connection between the actuator and the payload to reduce the power dissipated by the actuator due to friction. That is, the flexible connection needs to be designed to minimize the actuator's motion while maintaining the load's motion specified by $x_l(t) = X_d \sin(\omega_d t)$. The schematic of the flexible shaking device is shown in **Fig. 2**, in which K denotes the spring constant of the flexible connection.

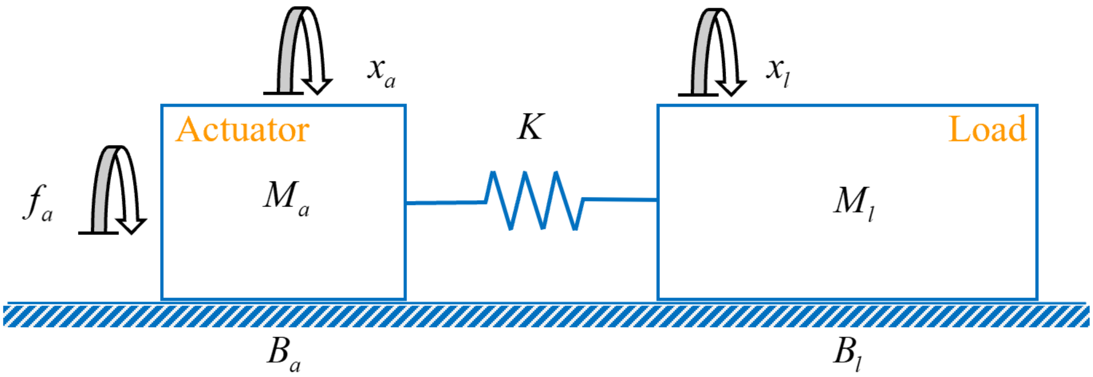

Fig. 2. Schematic of a shaking device with a flexible connection

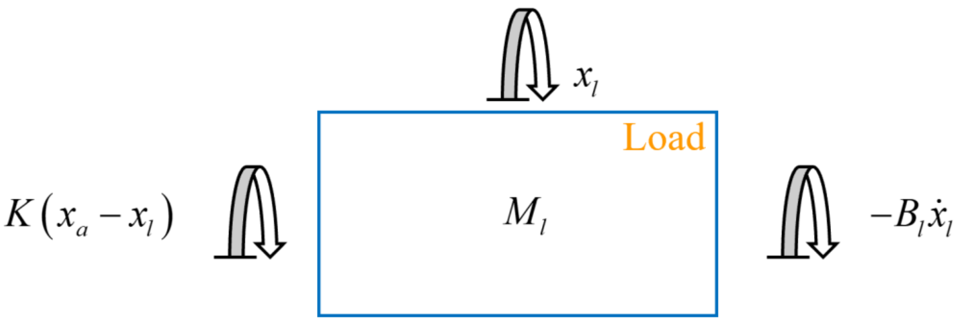

Fig. 3. Free-body diagram at the load side

Consider the free-body diagram at the load side as shown in **Fig. 3**. Applying Newton's second law of motion gives:

$$K\left(x_a - x_l\right) - B_l\dot{x}_l = M_l\ddot{x}_l, \text{ or} \tag{14}$$

$$M_l\ddot{x}_l + B_l\dot{x}_l + Kx_l = Kx_a. \tag{15}$$

Substituting $x_l(t) = X_d \sin(\omega_d t)$ into this equation gives:

$$-M_l X_d \omega_d^2 \sin(\omega_d t) + B_l X_d \omega_d \cos(\omega_d t) + KX_d \omega_d \sin(\omega_d t) = Kx_a, \text{ or} \tag{16}$$

$$\begin{aligned} x_a(t) &= \frac{X_d}{K}\left[B_l\omega_d \cos(\omega_d t) + \left(K - M_l\,\omega_d^2\right)\sin(\omega_d t)\right] \\ &= \frac{\sqrt{B_l^2\omega_d^2 + \left(K - M_l\,\omega_d^2\right)^2}}{K} X_d \sin(\omega_d t + \phi), \end{aligned} \tag{17}$$

in which $\phi = \tan^{-1}(B_l\omega_d / (K - M_l\omega_d^2))$. Therefore, the amplitude of the actuator's motion, denoted as X_a, is:

$$X_a = \frac{\sqrt{B_l^2\omega_d^2 + \left(K - M_l\,\omega_d^2\right)^2}}{K} X_d\,. \tag{18}$$

By equation (13), considering the case that $K \neq M_l\,\omega_d^2$, the power dissipated by viscous friction at the actuator side is:

$$P_f = B_a\dot{x}_a\dot{x}_a, \text{ where } \dot{x}_a(t) = X_a\omega_d \cos(\omega_d t + \phi). \tag{19}$$

The average power dissipation is:

$$P_f^{avg} = \frac{1}{2}B_a\left(\frac{B_l^2\omega_d^2 + \left(K - M_l\,\omega_d^2\right)^2}{K^2}\right)X_d{}^2\omega_d^2. \tag{20}$$

Equation (20) can also be expressed by:

$$P_f^{avg} = \frac{1}{2}B_a\left(\left(\frac{B_l\,\omega_d}{K}\right)^2 + \left(1 - \frac{M_l\,\omega_d^2}{K}\right)^2\right)X_d{}^2\omega_d^2. \tag{21}$$

Substituting K by equations (10) and (11) gives:

$$P_f^{avg} = \frac{1}{2} B_a \left(\left(\frac{(B_l\, \omega_d)(32l)}{(G\pi d^4)} \right)^2 + \left(1 - \frac{(M_l\, \omega_d^2)(32l)}{(G\pi d^4)} \right)^2 \right) X_d{}^2 \omega_d^2. \tag{22}$$

Equation (22) expresses the average power dissipation of the viscous friction at the actuator side.

To minimize X_a, choose $K = M_l\, \omega_d^2$, giving the minimum amplitude of the actuator's motion, $X_a = \frac{B_l\, \omega_d}{K} X_d = \frac{B_l}{M_l\, \omega_d} X_d$. With this minimum amplitude of the actuator's motion, the average power dissipated by viscous friction at the actuator side is:

$$P_{f\ \mathrm{Optimal}}^{\mathrm{avg}} = \frac{1}{2} \left(\frac{B_l}{M_l\, \omega_d} \right)^2 B_a X_d^2 \omega_d^2. \tag{23}$$

If $B_l < M_l\, \omega_d$, one has $X_a < X_d$, meaning that resonance occurs. More importantly, $P_f^{\mathrm{avg}} < P_r^{\mathrm{avg}}$, thus saving the power dissipated at the actuator side.

Remark 1: If $B_l = 0$, then equation (23) gives $X_a = 0$, meaning that the actuator does not have to move. Since there is zero power dissipated by viscous friction at the load side, the actuator does not have to move to provide power to the load. Hence, $P_{f\ \mathrm{Optimal}}^{\mathrm{avg}} = 0$.

B. Variables

Table 1 Variables of this study

	Variable	Justification/Methods
Independent variable	The length of the stainless steel rod. Lengths of 0m, 0.055m, 0.105m, 0.172m, and 0.253m.	Use stainless steel rods of various lengths and measure their lengths by using a ruler.
Controlled variable	X_d, amplitude of load oscillations.	Control the output amplitude by adjusting the input voltage's amplitude.
	ω_d, the oscillating frequency.	Control the frequency by controlling the input voltage's frequency in every trial.
	M_l at the load side, M_a at the actuator side.	Control the moment of inertia of the actuator and load by using the same equipment in every trial.
	v_s of the circuit.	Control the voltage by using the same value in every trial.
	B_a and B_l for the viscosity coefficient.	Control the constants by using the same equipment in every trial.
	d, diameter of the rod	Use the same type of bicycle spoke.
Dependent variable	Average power dissipated by the whole system.	Measured by the average power supplied by a DC power supply.

C. Materials

1. Controller Board (ESP32)
2. Oscilloscope
3. DC Motor (Actuator)
4. Load (Another DC motor acts as the load)
5. Power Supply
6. Current Probe (Tektronix A622)
7. Stands
8. USB Connection Cable
9. Wires

10. Multimeter
11. Breadboard
12. Stainless Steel Rod (Bicycle Spoke)
13. DC Motor Driver
14. Scope Probe
15. USB flash drive
16. Ruler (not shown in **Fig.4** and **6**)

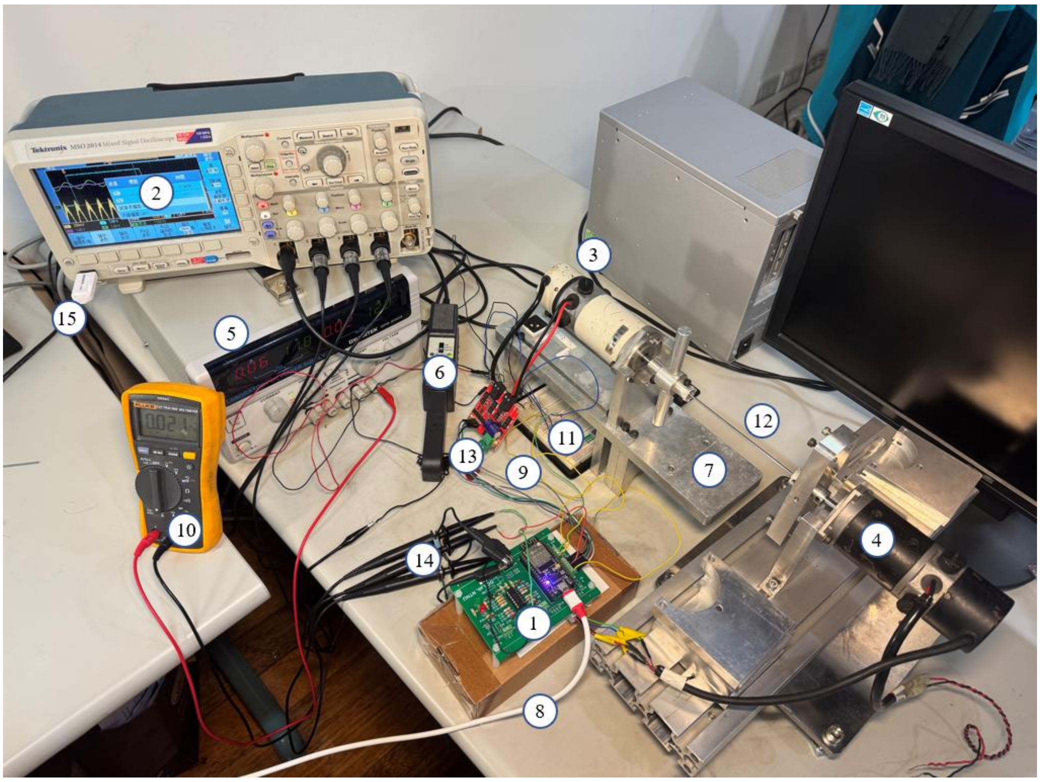

Fig. 4 Photo of the experimental setup

D. Procedure

Fig. 5 Block diagram of procedure

a. Experiment Setup

Fig. 6 illustrates the schematic of the experimental setup, including all the necessary materials and the position of the equipment.

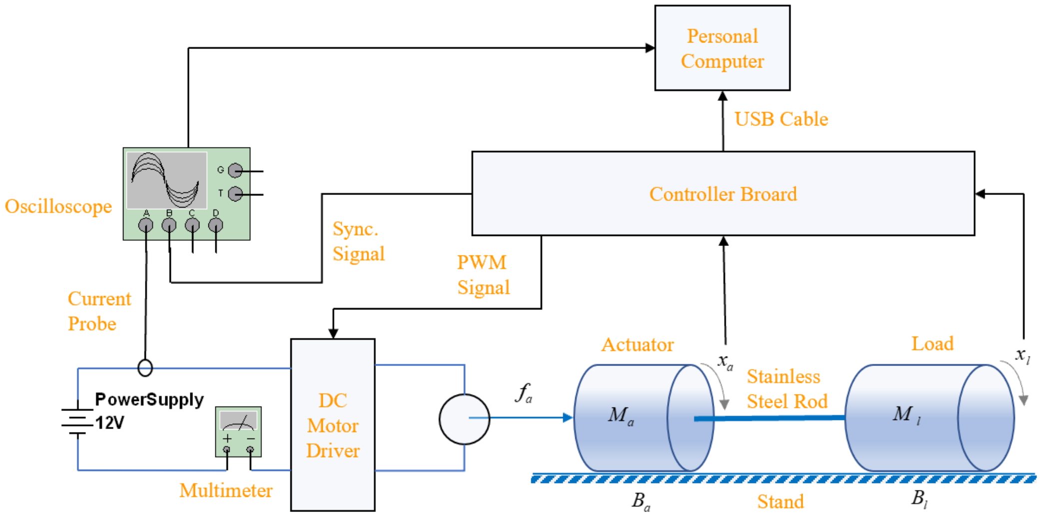

Fig. 6 Schematic of the experimental setup

b. Experimentation

1. Measure and record the length of the stainless steel rod.
2. Input the desired amplitude of the input voltage and the desired input oscillation frequency by the personal computer (PC). These data will be transmitted to the ESP32 through a USB cable. The ESP32 outputs a PWM signal to the DC motor driver that drives the actuator.
3. Press a preset button on the controller board to stop the operation.
4. The data will be transmitted through the USB cable to the PC and shown on the screen of the personal computer.
5. Press the "Save" button on the oscilloscope to record the input current. The data of the input current will be recorded into a USB flash drive attached to the oscilloscope.
6. After recording the input current, press the "Single" button on the oscilloscope to clear the previous data for the next trial.
7. Repeat all the steps for different trials.
8. After recording 5 trials for the same rod length, change to another stainless steel rod for the following lengths: {0m, 0.055m, 0.105m, 0.172m, 0.253m}.

c. Data Analysis

1. Combine the data from the USB drive and the data from the controller board.
2. Import the data of the oscillation and the input current to an application software for mathematical analysis, called MATLAB.
3. Calculate the average input power and load output amplitude in MATLAB.
4. Create graphs for further analysis.

V.Data Collection

Fig. 7 presents data of one trial for l=0.172m. The output amplitude is calculated as half the difference between the maximum and minimum values of the load position, derived from the two data tips in the figure. The current is measured using a probe, and its average value is taken.

L=17.2cm   Current = 0.3139A   Output Amplitude = 12.87°

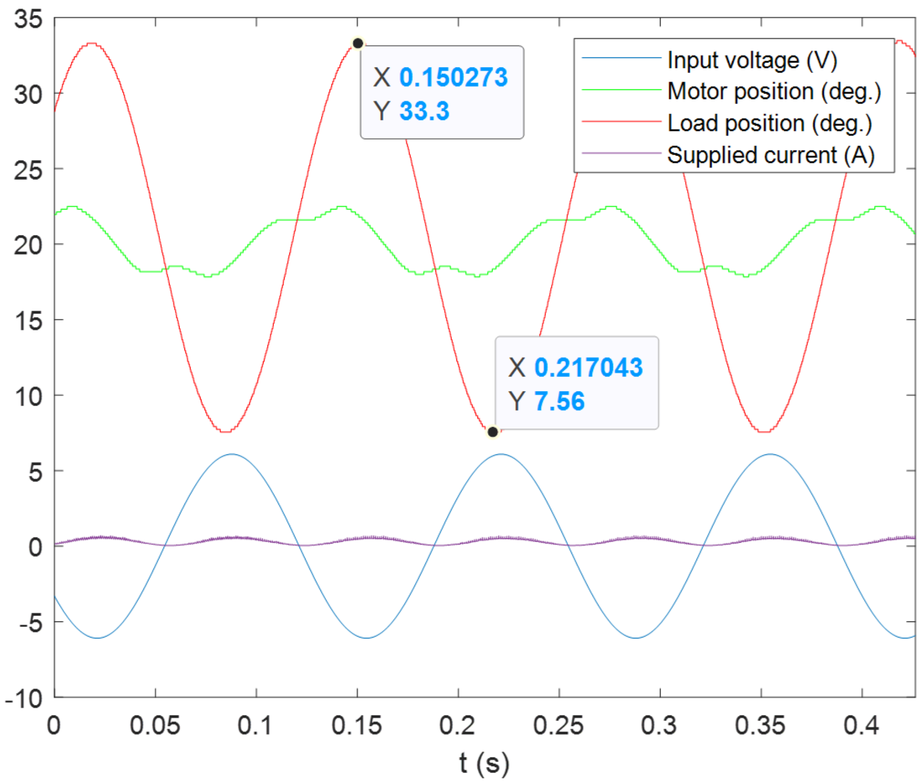

Table 2 Standard values of the experimentation

Input Voltage / V	Input Oscillation Frequency / Hz	Input Angular Frequency (ω_d) / rad·s^{-1}	Diameter of the Rod / m	Average Load Output Amplitude (X_d) / °
12±0.1	7.5±0.1	47.1±0.1	0.00200±0.00001	12.87±0.01

Table 3 Experimental result of rod-length 1

Length of Stainless Steel Rod (*l*)/ m			
l=0			
	Average Input Current / A	Average Input Power / W	Load Output Amplitude / °
Trial 1	0.3681±0.0001	4.42±0.04	12.51±0.01
Trial 2	0.3702±0.0001	4.44±0.04	12.96±0.01
Trial 3	0.3686±0.0001	4.42±0.04	12.87±0.01
Trial 4	0.3710±0.0001	4.45±0.04	12.96±0.01
Trial 5	0.3699±0.0001	4.44±0.04	13.05±0.01
Average	**0.3696±0.0001**	**4.44±0.04**	**12.87±0.01**

Table 4 Experimental result of rod-length 2

Length of Stainless Steel Rod (*l*)/ m			
l=0.055±0.001			
	Average Input Current / A	Average Input Power / W	Load Output Amplitude / °
Trial 1	0.3570±0.0001	4.28±0.04	13.41±0.01
Trial 2	0.3601±0.0001	4.32±0.04	12.96±0.01
Trial 3	0.3552±0.0001	4.26±0.04	12.60±0.01
Trial 4	0.3577±0.0001	4.29±0.04	12.60±0.01
Trial 5	0.3594±0.0001	4.31±0.04	12.60±0.01
Average	**0.3579±0.0001**	**4.29±0.04**	**12.83±0.01**

Table 5 Experimental result of rod-length 3

Length of Stainless Steel Rod (*l*)/ m			
l=0.105±0.001			
	Average Input Current / A	Average Input Power / W	Load Output Amplitude / °
Trial 1	0.3356±0.0001	4.03±0.03	12.60±0.01
Trial 2	0.3390±0.0001	4.07±0.04	12.78±0.01
Trial 3	0.3397±0.0001	4.08±0.04	12.78±0.01
Trial 4	0.3400±0.0001	4.08±0.04	12.78±0.01
Trial 5	0.3379±0.0001	4.05±0.03	12.78±0.01
Average	**0.3384±0.0001**	**4.06±0.04**	**12.74±0.01**

Table 6 Experimental result of rod-length 4

Length of Stainless Steel Rod (l)/ m			
l=0.172±0.001			
	Average Input Current / A	Average Input Power / W	Load Output Amplitude / °
Trial 1	0.3119±0.0001	3.74±0.03	12.78±0.01
Trial 2	0.3123±0.0001	3.75±0.03	12.69±0.01
Trial 3	0.3154±0.0001	3.78±0.03	12.87±0.01
Trial 4	0.3102±0.0001	3.72±0.03	12.69±0.01
Trial 5	0.3139±0.0001	3.77±0.03	12.87±0.01
Average	**0.3127±0.0001**	**3.75±0.03**	**12.78±0.01**

Table 7 Experimental result of rod-length 5

Length of Stainless Steel Rod (l)/ m			
l =0.253±0.001			
	Average Input Current / A	Average Input Power / W	Load Output Amplitude / °
Trial 1	0.2769±0.0001	3.32±0.03	13.05±0.01
Trial 2	0.2793±0.0001	3.35±0.03	13.14±0.01
Trial 3	0.2657±0.0001	3.19±0.03	12.42±0.01
Trial 4	0.2832±0.0001	3.40±0.03	13.41±0.01
Trial 5	0.2839±0.0001	3.40±0.03	13.50±0.01
Average	**0.2778±0.0001**	**3.33±0.03**	**13.10±0.01**

Table 8 Average Experimental result of stainless-steel rods

Length of Stainless Steel Rod ±0.001 /m	Average Input Current ±0.0001 / A	Average Input Power / W	Load Output Amplitude ±0.01 / °
0	0.3696	4.44±0.04	12.8700
0.055	0.3579	4.29±0.04	12.8340
0.105	0.3384	4.06±0.04	12.7440
0.172	0.3127	3.75±0.03	12.7800
0.253	0.2778	3.33±0.03	13.1040

Note 1: The following analysis will use the average results of each length.

In **Table 8**, the trend suggests that the power consumption decreases as the length of the rod increases, which matches the hypothesis from equation (22).

VI. Data Analysis

A. Length and Power Consumption

In this section, the discussion will focus on the relationship between power consumption and the amplitude of load oscillations with various lengths.

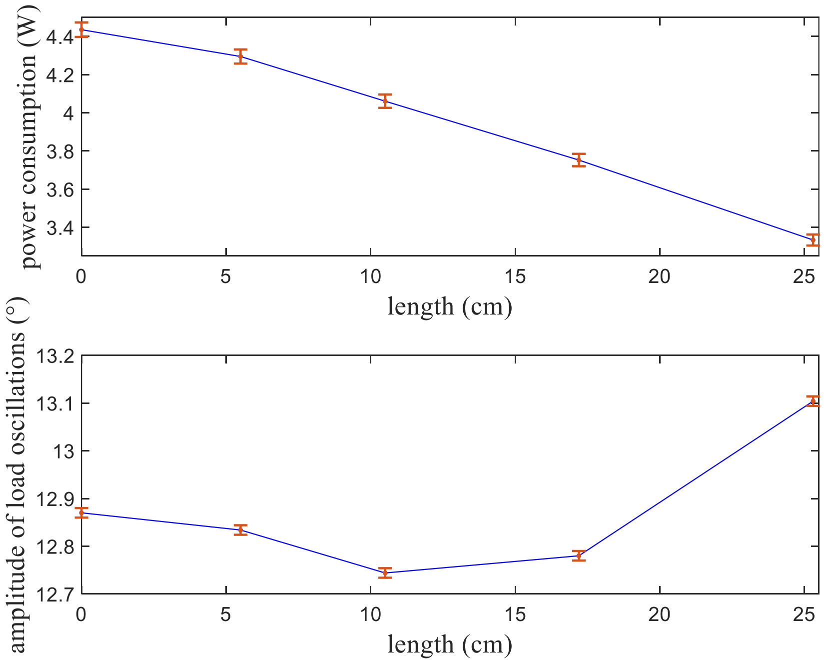

Fig. 8 Relationship between power consumption percentage and amplitude of load oscillations percentage with length

In **Fig. 8** (top), power consumption is expressed as a percentage to illustrate relative loss, with 100% corresponding to 4.4 W, which is the power consumption when l=0. Due to resonance in the stainless steel rod, the power consumption decreases by 24.8% when l=0.253. In **Fig. 8** (bottom), the "100%" amplitude is 12.87°, which is the amplitude of load oscillations when l=0. Although amplitude is a controlled variable, precise control is challenging, leading to a variation of 0.36°, or approximately 2.80%, which remains within an acceptable range for analysis. Considering both subplots, the actual percentage power consumption reduction when l=0.253 exceeds 25% since its amplitude is larger than the amplitude when l=0. This confirms that resonance significantly reduces the actuator's power dissipation, addressing the research question.

B. Spring Constant and Optimal Solution

In the references (*Bicyclewheel.info*, 2018) and (The Engineering ToolBox, 2007), Young's modulus and Poisson's ratio of stainless steel are 210 GPa and 0.3. From equations (10) and (11), the shear modulus can be expressed by:

$$G = \frac{E}{2(1+v)}, \quad G = \frac{210\times10^{9}}{2(1+0.3)}. \tag{24}$$

From the calculation, the shear modulus is 80.7692 GPa. Therefore, the spring constant of the stainless steel rod can be calculated by:

$$K = \frac{\left(80.7692\cdot10^{9}\right)\pi\left(0.0020\right)^{4}}{32l}. \tag{25}$$

For different lengths, **Table 9** shows the spring constant of the stainless steel rod with respect to the length of the rod.

Table 9 Spring constant of the stainless steel rod

Length (l) / m	Spring constant (K) / Nm^{-1}
l=0	Undefined
l=0.055±0.001	2.30±0.02
l=0.105±0.001	1.21±0.02
l=0.172±0.001	0.75±0.03
l=0.253±0.001	0.50±0.05

In **Theoretical Analysis**, the average power dissipation is expressed by equation (22). Since B_a, B_l, ω_d, M_l, and X_d are constant, an estimated trendline function can be expressed by:

$$P_f^{avg}(K) = a\left(\left(\frac{b}{K}\right)^2 + \left(1-\frac{c}{K}\right)^2\right), \tag{26}$$

where a, b, c are constants that can be determined by plotting the power consumption against the spring constant, which is shown in **Fig. 9**.

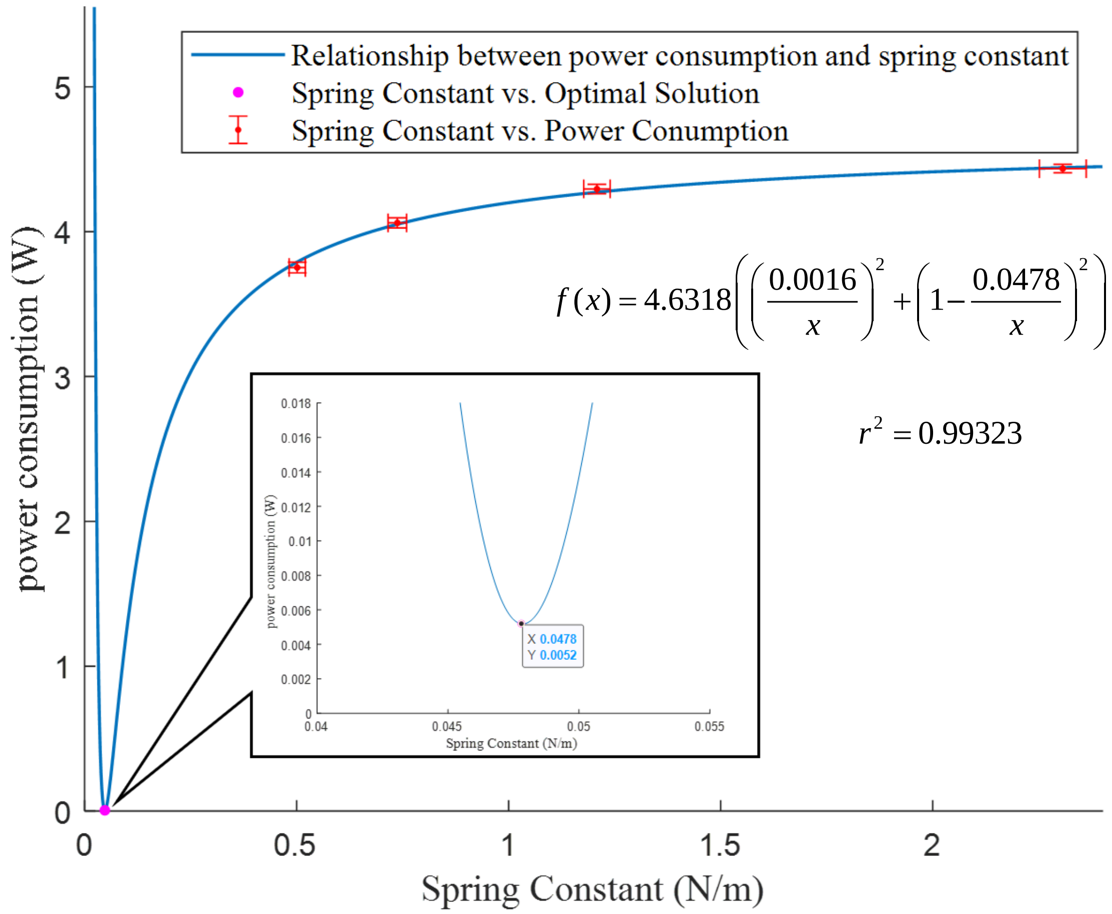

Fig. 9 Relationship between power consumption and spring constant

Fig. 9 shows the trendline equation expressed by equation (26). The insert of **Fig. 9** shows the optimal solution when $K = M_l \omega_d^2 = 0.0478$. The function's asymptote is $y = 4.6318$, representing the theoretical power consumption when l=0. Experimentally, when l=0, the power consumption is 4.4352 W. The difference between the theoretical and experimental values, 0.1966 W, approximately 2% of the original value. Therefore, it is reasonable to conclude that the equation of the trendline and the experimental results are compatible.

The constants B_a, B_l, ω_d, M_l, and X_d can solve for the optimal solution by equation (23). Since ω_d and X_d is known, the values of B_a, B_l and M_l can be calculated by the trendline equation (22) and equation (26), which gives:

$$4.6318\left(\left(\frac{0.0016}{x}\right)^2+\left(1-\frac{0.0478}{x}\right)^2\right)=\frac{1}{2}B_a\left(\left(\frac{B_l\,\omega_d}{K}\right)^2+\left(1-\frac{M_l\,\omega_d^2}{K}\right)^2\right)X_d{}^2\omega_d^2. \tag{27}$$

For simplification, equation (27) can be broken down into the following equations:

$$\begin{aligned} B_l &= \frac{0.0016}{\omega_d}, \\ M_l &= \frac{0.0478}{\omega_d^2}, \\ B_a &= \frac{2\times 4.6318}{X_d{}^2\omega_d^2}. \end{aligned} \tag{28}$$

Constants B_a, B_l, and M_l have the following values:

$$\begin{aligned} B_l &= 3.3953\times 10^{-5}, \\ M_l &= 2.1525\times 10^{-5}, \\ B_a &= 2.5199\times 10^{-5}. \end{aligned} \tag{29}$$

When $K = M_l\,\omega_d^2$, the optimal solution can be determined by equation (23), giving:

$$P_{f\ \text{Optimal}}^{\text{avg}} = \frac{1}{2}\left(2.5199\times 10^{-5}\right)\left(\frac{\left(3.3953\times 10^{-5}\right)}{\left(2.1525\times 10^{-5}\right)\left(47.1239\right)}\right)^2\left(12.8664\right)^2\left(47.1239\right)^2. \tag{30}$$

That is, when $K = M_l\,\omega_d^2 = 0.0478$, the optimal solution gives:

$$P_{f\ \text{Optimal}}^{\text{avg}} = 0.0052\ \text{W}. \tag{31}$$

The insert of **Fig. 9** shows the optimal solution when $K = M_l\,\omega_d^2 = 0.0478$, which is a rod length of 2.65 m. However, it far exceeds the longest value tested in this experiment, 0.253m, that gives a power dissipation of 3.33 W, which is about is about 75% of the original power dissipation. The theoretical minimum power dissipation is 0.0052 W, which is about 0.117% of the original value. However, a 2.65 m long rod would be impractical for a real-world application. Plus, other factors would become significant, like the bending of the rod under its weight, which wasn't accounted for in the theoretical analysis. Alternatively, decreasing the diameter could be another approach. Given that the diameter is raised to the power of four in equation (10), a slight decrease can significantly reduce the spring constant. However, this makes it more susceptible to bending or breaking.

VII. Evaluation

A. Percentage Uncertainty and Percentage Error

The comparison of percentage uncertainty and percentage error will determine whether there are random errors or systematic errors in this experiment. The percentage error can be written as:

$$\delta_{\text{Percentage Error}} = \left[\frac{|v_T - v_E|}{v_T}\right] \times 100, \tag{32}$$

where $\delta_{\text{percentage error}}$ is the percentage error, v_T and v_E are the theoretical value and the experimental value of the power consumption.

Table 7 Percentage uncertainty and percentage error

Length (l) / m	0	0.055	0.105	0.172	0.253
Spring constant (K) / Nm^{-1}	Undefined	2.3068	1.2083	0.7376	0.5015
Theoretical power dissipation / W	4.6318	4.4418	4.2726	4.0509	3.7910
Experimental power dissipation / W	4.4352	4.2948	4.0608	3.7524	3.3336
Percentage uncertainty	0.8604%	0.8613%	0.8629%	0.8653%	0.8693%
Percentage error	4.2446%	3.3095%	4.9572%	7.3687%	12.0654%

Note 2: The theoretical power dissipation is calculated by the trendline equation (23).

The average percentage uncertainty and error are calculated by the following equations:

$$\bar{\delta}_{\text{Percentage Uncertainty}} = \frac{0.8604 + 0.8613 + 0.8629 + 0.8653 + 0.8693}{5} = 0.8638\%, \quad (33)$$

$$\bar{\delta}_{\text{Percentage Error}} = \frac{4.2446 + 3.3095 + 4.9572 + 7.3687 + 12.0654}{5} = 6.3891\%. \quad (34)$$

The average percentage error exceeds the average percentage uncertainty, indicating that systematic errors play a more significant role in the experiment.

Systematic Errors:

1. The resistance of the wires will affect the power dissipation.
2. Friction between the actuator and the load.
3. The viscous force between the connection hole and the stainless steel will affect the output load amplitude.
4. When l=0, super glue is used to connect the actuator with the load. The results show that the viscous force of the super glue does not provide a rigid body connection as the theoretical and experimental values are different.
5. The output load amplitudes for all cases should be constant to make a more applicable analysis of the relationship between power dissipation and length.

Random Errors:

1. Unstable viscous force in the connection hole.
2. Variability in the glue's bonding strength.
3. When the actuator is operating, the torsion might make the rod unable to align the two connectors in a straight line.

B. Limitations

Primarily, as mentioned in **Spring Constant and Optimal Solution**, while the trendline aligns with the experimental results, the theoretical extremum has not been tested experimentally.

Secondly, the theoretical power consumption derived in the **Theoretical Analysis** considers only the actuator's power consumption. However, the data on power consumption in **Data Collection** represents the whole system's power consumption, including losses from the motor driver, wire resistance, and load's friction, which could be the origin of the discrepancy for the percentage error. These factors unaccounted for in the **Theoretical Analysis** are unknown, indicating the experiment's most significant uncertainty. For further analysis, their effects are assumed to be negligible. However, it is crucial to outline these uncertainties.

C. Improvements

The following ways can improve the experiment's accuracy and reliability. Primarily, use a stronger, rigid adhesive to ensure a more stable connection when l=0. Secondly, implement active feedback control to maintain a constant output amplitude to enhance the accuracy of the analysis when comparing each data point. In terms of random errors, use a controlled amount of lubricant with consistent viscosity to minimize variations in resistance. Furthermore, using longer and precision-machined slots and rails can help keep the connectors in a straight line, so that the rod can align the two connectors in a straight line.

Based on the available materials for the experimentation, the possible way to test the extremum is to alter the input oscillation frequency. Since the value of K when l=0.253 is known and assuming that M_l remains the same, the optimal solution can be calculated by $K = M_l \omega_d^2$. Although M_l is a constant, it has to be determined experimentally. In this case, when ω_d =152.64 rad/s or 24.29 Hz, the rod with a length of 0.253 m will present an optimal solution.

D. Extension

An extension could involve testing power consumption versus spring constant while keeping the length constant and varying the thickness. To further improve power efficiency, a capacitor can be added in parallel between the power source and the actuator circuit, as shown in **Fig. 10**. That is, a capacitor, C, is added to reduce the power dissipated by the power source's internal resistance, R_s . If the RLC circuit is operating at resonance, the capacitor, C, supplies electric power to the actuator so that the amplitude of i_s is reduced. As a result, the power dissipated by the power source's internal resistance, R_s, is diminished.

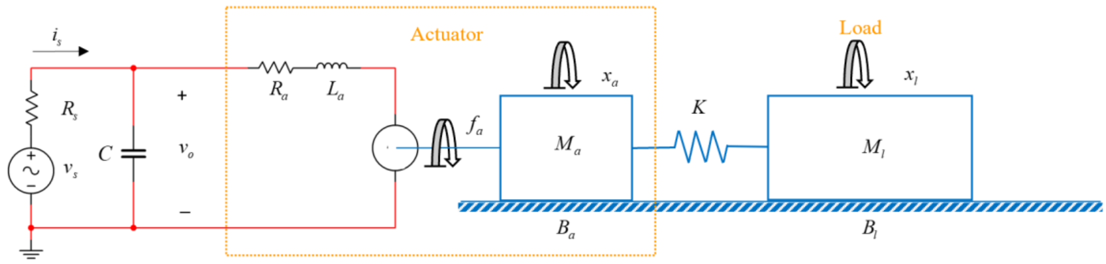

Fig. 10 Schematic of a capacitor-added system

VIII. Conclusion

In conclusion, this study investigates the relationship between power dissipation and the length of a stainless steel rod. Resonance is the key concept in the reduction of power dissipation of the system. The experimental results verify that the relationship can be expressed by equation (27), $P=4.6318\left(\left(\frac{0.0016}{K}\right)^2+\left(1-\frac{0.0478}{K}\right)^2\right)$, where the spring constant is inversely proportional to the rod's length. Leveraging resonance reduced power dissipation to 75% of its original value, with an optimal dissipation of 0.0052 W, or 0.117% of the original.

However, the experimental optimal solution is needed to comprehensively prove that the **Theoretical Analysis** is suitable in all cases. Beyond increasing length, reducing the rod's diameter effectively lowers the value of the spring constant, but excessive thinning increases the risk of breakage.

Ultimately, the analysis of this study answers the research question "*How does power dissipation in a shaking device respond to changes in the spring constant caused by altering the length of the stainless steel rod (connection between actuator and load)?*" Although the study faced a few limitations, it paved the way for further exploration in the resonance and mass-spring-damper systems. By understanding the motion of the system and addressing the challenges of the errors and limitations, this study lays the groundwork for further advancements in this field.

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